v4 check — gate failed, no tiles

Rule B (`own_range_grid`) on the buildings-only surface, ground replaced by the AHN4 DTM (`dtm_05m`). Known, non-water pixels. No 30 m clutter dilation. P is `20190822074237_20190822074521`. K is `20190823072627_20190823072925`.

The layover gate is +1.5 dB in the near and middle thirds on **both** strips. P misses it. K passes. No tiles were built. `train.py` was not run.

1. Old ground minus DTM

5th-percentile local ground minus the DTM, non-water land, both finite.

strip	n	median	p10	p90
P	8,847,273	−0.34 m	−3.19 m	+0.07 m
K	8,018,479	−0.42 m	−2.78 m	+0.05 m

The median is a few tens of centimetres (the old ground is slightly lower). The low tail is about 3 m. That is the sunken-street / water-edge bias the DTM removes. It is not a uniform height shift.

2a. B-bldg contrast

Median HH minus the nominal median. dB.

strip	third	layover	shadow	shadow interior (eroded 2 m)	n layover	n nominal
P	near	+1.18	−0.88	−1.76	6,159,319	20,201,097
P	middle	+1.04	−1.14	−1.62	4,451,671	34,102,977
P	far	+0.78	−0.58	−1.22	1,055,023	25,177,975
P	all	+1.22	−0.88	−1.52	11,666,013	79,482,049
K	near	+2.16	−1.20	−1.08	9,643,915	18,146,885
K	middle	+2.62	−1.18	−1.34	12,186,479	28,393,635
K	far	+3.10	−1.28	−1.66	7,257,122	20,955,677
K	all	+2.58	−1.24	−1.48	29,087,516	67,496,197

P fails both gated thirds. K clears both. Same code, same DTM, so this is the strip, not a flipped sensor: K's alpha still falls toward the south, and the labels are written north-up.

This is not the rule-check number. That gate was scored on the pixels known in all five label sets and outside the 30 m clutter buffer, about 3% of the port strip, where B-bldg layover was +1.94 / +3.64 dB. Here every known dry pixel is in the median, including the dim tail of the layover class. That is the mask this check asked for, and on P it is under +1.5 dB.

2b. Shadow interior

Eroding the shadow class by 2 m makes it darker on P (interior −1.52 against the class −0.88) and slightly darker overall on K (−1.48 against −1.24). K's near third goes the other way (−1.08 interior, −1.20 for the class). Neither interior reaches the open-water floor from DECIDE (about −1.9 to −2.1). Reported only.

2c. Roof test

Pixels inside the footprint, eroded 2 m, at least `6 m + max(0, s) / tan θ` behind the near wall. Buildings with nDSM p90 ≥ 8 m and at least 100 pixels in both the layover part (coverage ≥ 1.5) and the nominal part. The number is median HH(layover) − median HH(nominal), then the median over buildings.

	n	median dB	fraction > 0
P all	55	+0.50	0.67
P 8–15	23	+0.52	0.74
P 15–30	27	+0.50	0.67
P > 30	5	−0.54	0.40

	n	median dB	fraction > 0
K all	333	+0.95	0.71
K 8–15	204	+1.10	0.74
K 15–30	105	+0.91	0.67
K > 30	24	+0.78	0.63
pooled	388	+0.87	0.70

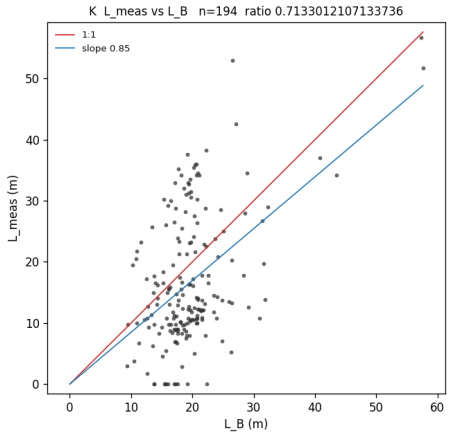
Pooled median is +0.87 dB and 70% of buildings are positive. That is between the two bars (+1.0 with fraction ≥ 0.6, or below +0.5). The rule says keep rule B and say so. It does not justify the roof-nominal edit. It also does not override the failed brightness gate.

2d. Layover length on K, DTM ground

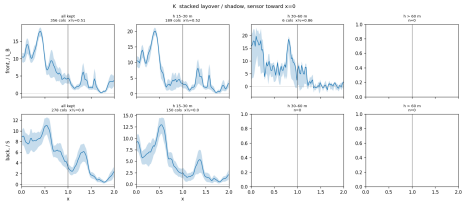
Same column rule as DECIDE.md Part 2. Kept 386 columns, 208 buildings (was 293 / 172 with the 5th-percentile ground). Scatter n = 194.

	DTM	DECIDE (old ground)
median L_meas / L_B	0.71	0.77
slope through the origin	0.85	0.89
RMS vs 1:1	10.0 m	10.0 m
median L_B	19.0 m	18.5 m
median L_meas	13.6 m	13.8 m
median L_meas / L_A	1.07	1.14
half-power x, all	0.51	0.57
back recover x	0.0	0.0

The one clean tall column is unchanged: `0599100000750035`, h 52 m, L_B 57.3 m, L_meas 56.8 m. Half-power in the 30–60 m bin is 0.86 but that bin is 6 columns. Nothing over 60 m survived the isolation cut.



K, DTM ground: measured layover length against L_B. The line through the origin has slope 0.85.



K stacked profiles. Front half-power is at 0.51 L_B. The back side is already at the background.

Stop

Do not tile. Do not train. A v4 built on this mask would put a layover class on P whose median is only about +1.1 dB above nominal in the gated thirds. K would have passed on its own.